Data Understanding

Phase 03 — CRISP-DM Methodology

Retail Business Intelligence Platform

1. Introduction

The **Data Understanding** phase represents the second technical step of the **CRISP-DM (Cross Industry Standard Process for Data Mining)** methodology.

The main objective of this phase is to explore, analyze, and understand the available data before starting data cleaning, transformation, and integration processes into a Business Intelligence system.

This phase aims to:

- Identify the different data sources.
- Understand the structure of available datasets.
- Analyze data attributes and their meanings.
- Evaluate data quality.
- Identify relationships between business entities.
- Determine the required data for building the Data Warehouse and Power BI dashboards.

In the **Olist BI Analytics Project**, this analysis focuses on an e-commerce dataset containing information about customers, orders, products, sellers, payments, and customer reviews.

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2. Dataset Overview

2.1 General Description

The dataset used in this project is:

Brazilian E-Commerce Public Dataset by Olist

It represents real transactional data from a Brazilian e-commerce marketplace.

The dataset covers the period from **2016 to 2018** and provides information about:

- Customer behavior.
- Sales performance.
- Product transactions.
- Payment methods.
- Customer satisfaction.
- Seller performance.

2.2 Analysis Objectives

The main objectives of this analysis are:

- Understand the complete order lifecycle.
- Identify key business indicators (KPIs).
- Prepare data for the Data Warehouse.
- Build decision-making dashboards using Power BI.
- Prepare data for advanced analytics and predictive models.

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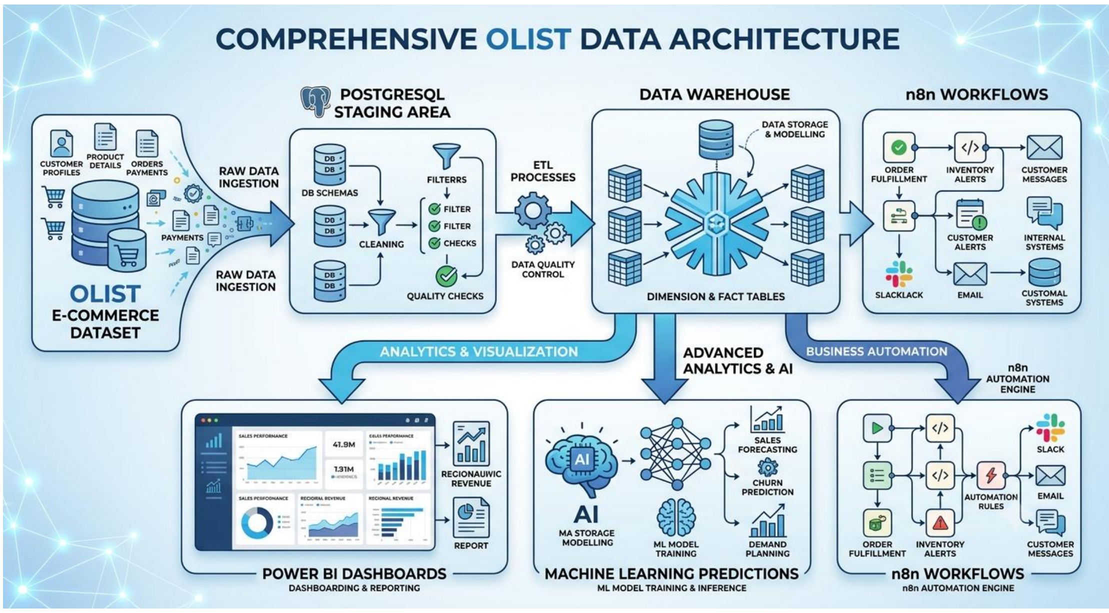
3. Data Sources

The Olist dataset is composed of multiple CSV files representing different business entities of the e-commerce platform.



4. Dataset Tables Description

4.1 Customers Table

Description

The Customers table contains information about platform users.

Principales colonnes :

Column	Description
customer_id	Customer identifier
customer_unique_id	Unique customer identifier
customer_city	Customer city
customer_state	Customer state

Business Role

This table represents the **Customer Dimension** in the future Data Warehouse.

It enables analysis of:

- Customer geographic distribution.
- Customer segmentation.
- Customer purchasing behavior.

4.2 Orders Table

Description

The Orders table represents the purchase transactions made by customers.

Column	Description
order_id	Order identifier
customer_id	Related customer
order_status	Order status
order_purchase_timestamp	Purchase date
order_delivered_customer_date	Delivery date

Business Role

This table represents the main sales process.

It allows analysis of:

- Order volume.
- Sales evolution.
- Delivery performance.
- Operational efficiency.

4.3 Order Items Table

Description

The Order Items table contains detailed information about products included in each order.

Column	Description
order_id	Order identifier
product_id	Product identifier
seller_id	Seller identifier
price	Product price
freight_value	Shipping cost

Business Role

This table represents the transactional sales details.

It allows analysis of:

- Revenue generation.
- Product performance.
- Seller performance.
- Sales volume.

4.4 Products Table

Description

The Products table contains information about available products.

Column	Description
product_id	Product identifier
product_category_name	Product category
product_weight_g	Product weight

Business Role

This table supports product and category performance analysis.

4.5 Payments Table

Description

The Payments table contains financial transaction information.

Column	Description
order_id	Order identifier
payment_type	Payment method
payment_value	Paid amount

Business Role

It allows analysis of:

- Revenue.
- Payment methods.
- Transaction values.

4.6 Reviews Table

Description

The Reviews table contains customer feedback and ratings.

Column	Description
review_score	Customer rating
review_comment_message	Customer comment
order_id	Related order

Business Role

It helps measure:

- Customer satisfaction.
- Service quality.
- Customer experience.

4.7 Sellers Table

Description

The Sellers table contains marketplace seller information.

Column	Description
seller_id	Seller identifier
seller_city	Seller city
seller_state	Seller state

Business Role

It supports seller performance analysis.

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5. Data Quality Analysis

An exploratory analysis was performed using **Python (Pandas)** to evaluate the quality of the dataset.

The following checks were performed:

5.1 Data Structure Analysis

For each table, the following elements were analyzed:

- Number of rows.
- Number of columns.
- Data types.
- General information.

Objective

Understand the structure of the datasets before loading them into the Data Warehouse.

5.2 Missing Values Analysis

Missing value analysis was performed to identify:

- Columns containing null values.
- Potential data quality issues.

These results will help define cleaning rules during the Data Preparation phase.

5.3 Duplicate Analysis

Duplicate detection was performed to verify:

- Identifier uniqueness.
- Data consistency.
- Data integrity.

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6. Business Tables Deep Analysis

The main business tables analyzed are:

- Orders.
- Order Items.
- Payments.
- Customers.

6.1 Orders Analysis

The Orders table analysis focused on:

- Order status distribution.
- Monthly order evolution.
- Delivery time analysis.

Business Insights

This analysis helps evaluate the operational performance of the order management process.

6.2 Order Items Analysis

The Order Items analysis focused on:

- Number of sold products.

- Product price distribution.
- Revenue generation.
- Best-selling products.

Business Insights

This table represents the main source for commercial KPIs.

6.3 Payments Analysis

The Payments analysis focused on:

- Payment method distribution.
- Revenue generated by payment type.
- Transaction value analysis.

Business Insights

These results will support financial KPI creation.

6.4 Customers Analysis

The Customers analysis focused on:

- Number of customers.
- Geographic distribution.
- Main customer locations.

Business Insights

This analysis supports the creation of the Customer Dimension in the Data Warehouse.

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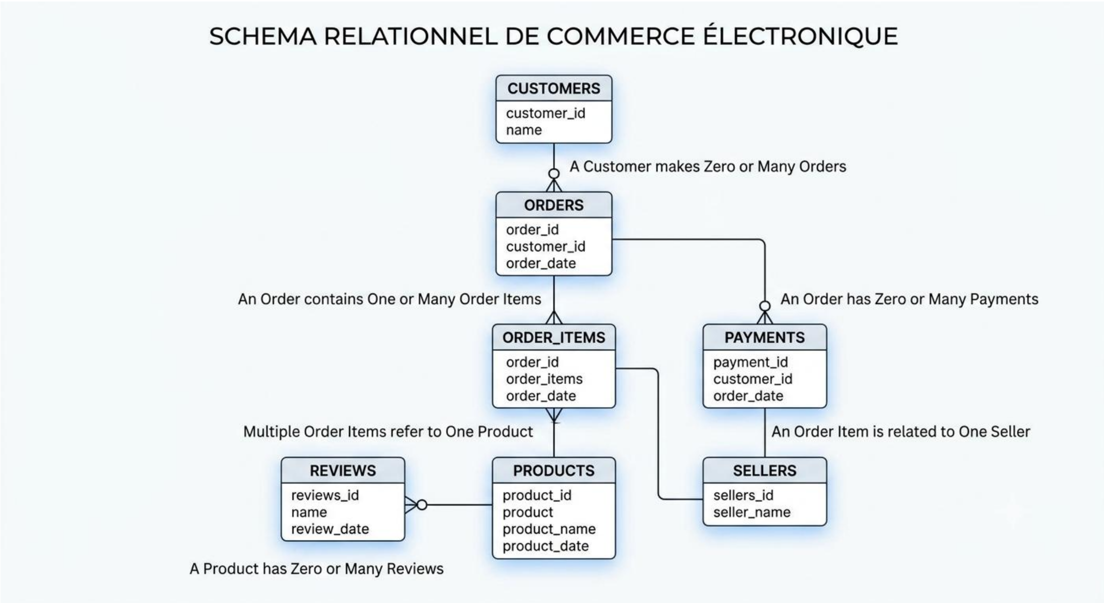
7. Data Relationships

The identified relational model is:



8. Summary of Findings

The Data Understanding phase allowed us to:

- Understand the Olist dataset structure.
- Identify the main business entities.
- Evaluate data quality.
- Analyze relationships between tables.
- Identify the required data for BI implementation.

The analyzed tables will be used during the next phase:

Data Preparation

which will include:

- Data loading into PostgreSQL.
- Data cleaning.
- Data transformation.
- Staging Area creation.

- Data Warehouse preparation.

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9. Conclusion

The Data Understanding phase is a critical step in the development of the **Olist BI Analytics Project**.

It provided a clear understanding of available data sources and prepared the foundation for building a complete Business Intelligence solution.

The final BI architecture will include:

- A Data Warehouse.
- Power BI dashboards.
- Machine Learning models.
- n8n automation workflows.